

Zirui (Ray) Chen

Krieger Hall 107, 3400 N Charles St.
Baltimore, MD 21218

Email: zchen160@jh.edu

EDUCATION

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| Johns Hopkins University
Ph.D. Cognitive Science
Advisor: Michael F. Bonner | 2023-present |
| Johns Hopkins University
M.A. Cognitive Science
Advisor: Michael F. Bonner | 2022-2023 |
| Emory University
B.S. Neuroscience & Behavioral Biology, B.A. Computer Science (GPA=3.94)
<i>Summa cum laude</i>
Honors thesis: The global color of a scene is relevant for human visual scene discrimination.
Advisor: Daniel D. Dilks | 2018-2022 |

PUBLICATIONS

- Chen, Z.**, Isik, L., & Bonner, M.F. (2026). Multidimensional dynamics of object representations in the human visual system. *bioRxiv*.
- Chen, Z.**, & Bonner, M.F. (2025). Universal dimensions of visual representation. *Science Advances*, 11, eadw7697.
- Cheng, A., **Chen, Z.**, & Dilks, D. D. (2023). A stimulus-driven approach reveals vertical luminance gradient as a stimulus feature that drives human cortical scene selectivity. *NeuroImage*, 269, 119935.

TALK PRESENTATIONS

- Chen, Z.**, & Bonner, M.F. (2023), Canonical Dimensions of Neural Visual Representation. *Annual Meeting of the Vision Sciences Society 2023*.
- Chen, Z.**, Cheng, A., & Dilks, D.D. (2020), Characterizing an image property relevant for human cortical scene processing. *Summer Undergraduate Research Experience (SURE) Virtual Symposium 2020*.

POSTER PRESENTATIONS

- Chen, Z.**, & Bonner, M.F. (2026), Revealing core representational axes of neural networks through learning dynamics and brain alignment. *Conference on Cognitive Computational Neuroscience 2026*.
- Chen, Z.**, & Bonner, M.F. (2026), Learning dynamics reveal the natural basis of visual representation. *Annual Meeting of the Vision Sciences Society 2026*.
- Chen, Z.**, Isik, L., & Bonner, M.F. (2025), Neural Dimensionality and Temporal Dynamics of Visual Representations. *Conference on Cognitive Computational Neuroscience 2025*.
- Chen, Z.**, Isik, L., & Bonner, M.F. (2025), Unraveling the geometry of neural representational dynamics in rapid visual processing. *Annual Meeting of the Vision Sciences Society 2025*.
- Chen, Z.**, & Bonner, M.F. (2024), Accounting for the reliability of deep neural networks in representational modeling. *Conference on Cognitive Computational Neuroscience 2024*.
- Chen, Z.**, & Bonner, M.F. (2024), How to estimate noise ceilings for computational models of visual cortex. *Annual Meeting of the Vision Sciences Society 2024*.

Chen, Z., & Bonner, M.F. (2023), Canonical Dimensions of Vision. *Conference of Cognitive Computational Neuroscience 2023*.

Elmoznino, E., **Chen, Z.**, & Bonner, M.F. (2022), Latent dimensionality scales with the performance of deep learning models of visual cortex. *Conference of Cognitive Computational Neuroscience 2022*.

Chen, Z., Cheng, A., & Dilks, D.D. (2021), Uncovering the visual features relevant to human visual scene discrimination. *Summer Undergraduate Research Experience (SURE) Virtual Symposium 2021*.

RELEVANT COURSEWORK

Computer Science: Analysis of Algorithms; Artificial Intelligence; Computational Linguistics; Data Structure and Algorithms; Machine Learning.

Mathematics: Differential Equations; Linear Algebra; Linear Optimization; Nonlinear Optimization; Multivariable Calculus; Numerical Analysis; Real Analysis.

Psychology/Neuroscience/Cognitive Science: Bayesian Inference; Behavioral Neuroscience; Brain Imaging; Cognition; Foundations of Cognitive Science; Neurobiology; Neuroeconomics: Decision Making; Perspectives in Neuroscience & Behavior; Predictive Modeling; Theory and Modeling of Information Coding in Neural Activity.

SKILLS

Programming languages: Python, MATLAB, R, Java, Bash, C, JavaScript, HTML/CSS, Swift.

Methods and tools: Machine learning, deep learning, fMRI pre-processing (FSL, fMRIPrep), image processing, experiment interface design (Psychtoolbox, MTurk, Prolific).

TEACHING EXPERIENCE

Teaching Assistant, Johns Hopkins University Course: Introduction to Cognitive Neuropsychology Professor: Michael McCloskey	2025
Teaching Assistant, Johns Hopkins University Course: Neuroscience: Cognitive Professor: Michael F. Bonner	2025
Teaching Assistant, Johns Hopkins University Course: Deep Learning for Cognitive Neuroscience Professor: Leyla Isik	2024
Teaching Assistant, Johns Hopkins University Course: Probabilistic Models of the Visual Cortex Professor: Alan Yuille	2023
Teaching Assistant, Emory University Course: Non-Linear Optimization Professor: Elizabeth Newman	2021
Teaching Assistant, Emory University Course: Computer Architecture/Machine Level Programming Professor: Shun Cheung	2021